GRADING SHEET

MILESTONE 1

5 _____ Cover page
– includes a conceptual figure of satellite(s)
- includes name of satellite on cover
- All group member's name
- Date and Course

5 _____ Introduction
– describe mission information
- overview of what will be in report

10 ____ 1) Establish a name for your satellite system and determine its mission objectives. Do not choose an orbit, however, until you have completed steps 2 – 4.
- State satellite system name and explain why you chose it
- Describe its mission objectives, both primary and secondary (if any)

10 ____ 2) Provide a summary of the Sun-Earth system, considering risks/hazards to satellite operations at two of the following GEO, MEO, or LEO. Quantify or characterize the different emissions from the Sun and the way they affect earth observation missions or damage spacecraft.
- Summary of Sun-Earth
    - charged particles – quantify or characterize
    - radiation – quantify or characterize
    - Earth's magnetic field
- Risks at *two* orbital altitudes – include at least three of the following:
    - charging
    - ESD
    - solar pressure
    - Van Allen radiation belts
    - communication interference

15 _____ 3) Provide a summary of what constitutes space weather, current research efforts or operational capabilities to measure space weather, and why your customer would want to be informed on space weather events. Include projected impacts to a communications downlink from either the GEO, MEO, or LEO solution to ground stations on the earth.
- Definition/explanation of space weather
- Research or operational ways to monitor it
- Why space weather is important/pertinent
- Impact to communications and/or other subsystems

GRADING SHEET

MILESTONE 1

10 _____ 4) Your customer is concerned about trying to minimize production costs for the satellite solution. One of the possible courses of action is to forgo vacuum testing for the space vehicles. Provide a summary of why vacuum testing would be advisable, considering risks/hazards to satellite operations at GEO, MEO, or LEO. Quantify or characterize the effects of a vacuum environment on mission accomplishment and potential damage to the spacecraft.

- Description of vacuum testing and its importance
- recommendation (should recommend to do vacuum testing unless perhaps cubesat mission)
- consideration of risk
- quantification or characterization

15 _____ 5) Based on your results, choose an orbit for your satellite or satellite constellation.

- Determine an appropriate orbit for your satellite
    - -- Orbital elements – side or altitude, eccentricity (or state circular), and inclination
    - -- Describe why you picked that orbit

5 ______ 6) Visual simulation

5 ______ 7) Orbital decay estimation

5 _____ Conclusions

5 _____ References

- In text citations used
- Good reference list in consistent format

10 _____ Grammar

- block justified
- flow
- formal paper guidelines
- few grammatical errors
- label figures and graphs